



Incident report analysis

Instructions

As you continue through this course, you may use this template to record your findings after completing an activity or to take notes on what you've learned about a specific tool or concept. You can also use this chart as a way to practice applying the NIST framework to different situations you encounter.

Summary	Organization recently experienced a DDoS attack, which compromised the internal network for two hours until it was resolved. This caused organization's network services suddenly stopped responding due to an incoming flood of ICMP packets. Later identified that a malicious actor had sent a flood of ICMP pings into the company's network through an unconfigured firewall. This vulnerability allowed the malicious attacker to overwhelm the company's network through a distributed denial of service (DDoS) attack.
Identify	The malicious attacker targeted the company with an ICMP flood attack to overwhelm the company's network through a distributed denial of service (DDoS) attack. The entire internal network was affected was affected due to this attack.
Protect	The network security team implemented, a new firewall rule to limit the rate of incoming ICMP packet, source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets, network monitoring software to detect abnormal traffic patterns, and an IDS/IPS system to filter out some ICMP traffic based on suspicious characteristics.
Detect	The cybersecurity team source IP address verification on the firewall to check for spoofed IP addresses on incoming ICMP packets and also put in place network monitoring software to detect abnormal traffic pattern.

Respond	For future security events, the cybersecurity team will isolate affected systems to prevent further disruption to the network. They will attempt to restore any critical systems and services that were disrupted by the event. Then, the team will analyzed firewall logs and IDS/IPS alerts to confirm the source and method of the attack. The team will also report all incidents to upper management and appropriate legal authorities, if applicable.
Recover	To recover from a DDoS attack by ICMP flooding, access to network services need to be restored to a normal functioning state. In the future, external ICMP flood attacks can be blocked at the firewall. Then, all non-critical network services should be stopped to reduce internal network traffic. Next, critical network services should be restored first. Finally, once the flood of ICMP packets have timed out, all non-critical network systems and services can be brought back online.

Reflections/Notes: Analyzing this cybersecurity attack scenario by using the NIST CSF was beneficial to create a quality cybersecurity incident report that can demonstrate a proactive approach to security, improving communication and transparency with stakeholders, and improve security practices within an organization. This helped me understand how potential attacks need to be viewed and approached.